Firm Age and Size in the Longitudinal Employer-Household Dynamics Data

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Abstract

The Census Bureau's Quarterly Workforce Dynamics (QWI) and OnTheMap now provide detailed workforce statistics by employer age and size. These data allow a first look at the demographics of workers at small and young businesses as well as detailed analysis of how hiring, turnover, and job creation/destruction vary throughout a firm's lifespan. Both the QWI and OnTheMap are tabulated from the Longitudinal Employer-Household Dynamics (LEHD) linked employer-employee data. Firm age and size information was added to the LEHD data through integration of Business Dynamics Statistics (BDS) microdata into the LEHD jobs frame. This paper describes how these two new firm characteristics were added to the microdata and how they are tabulated in QWI and OnTheMap.

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A. Overview

The Longitudinal Employer-Household Dynamics (LEHD) program recently added employer age and size to the Quarterly Workforce Indicators (QWI) and OnTheMap.[1] The Quarterly Workforce Indicators (QWI) provide local labor market statistics by industry, worker demographics, employer age, and employer size. Unlike statistics tabulated from firm or person-level data, the QWI source data are unique job-level data that link workers to their employers. Because of this link, labor market data in the QWI is available by both employer characteristics and worker characteristics such as age, sex, educational attainment, and race/ethnicity. This allows for analysis by demographics of a particular local labor market or industry – for instance, identifying industries with aging workforces. Links between workers and firms also allow the QWI to identify worker flows – hires, separations, and turnover – as well as net employment growth. Additionally, wages by industry and demographics and by whether the worker was newly hired are also available. The addition of firm age and size information expands the capacity of QWI to analyze worker demographics, hiring and turnover, job creation and wages, by the age and size of the firm.

The LEHD Origin-Destination Employment Statistics (LODES), more commonly referred to as OnTheMap, reports employment by both place of work and place of residence at block-level geography. The ability to analyze employment by both place of residence and place of work is critical for identifying regional labor markets and understanding the interconnectedness of geographic areas that lie across state and metro area boundaries. With the addition of firm age and size, OnTheMap users can select to display firm size and age classes, providing block level data on the pattern of start-up and small business activity in a local labor market.

[1] The initial release of QWI with firm age and size was spring of 2012. LODES/OnTheMap with firm age and size was first released in spring of 2013. Firm age and size have been added to the LEHD microdata (specifically, the Employer Characteristics File (ECF)) as described in this document. LEHD microdata files are available to researchers who have been approved for use of the files for research in the secure Census Research Data Centers.

The source data for the QWI and OnTheMap is the Longitudinal Employer-Household Dynamics (LEHD) linked employer-employee microdata. The LEHD data are a massive longitudinal database covering over 95% of U.S. private sector jobs. Much of this data are collected via a unique federal-state data sharing collaboration, the Local Employment Dynamics (LED) partnership. The LED partnership is a cooperative venture between the LEHD program at the U.S. Census Bureau and state agencies of all 50 states, the District of Columbia, Puerto Rico, and the U.S. Virgin Islands.[2] Partner states voluntarily provide data from existing administrative record systems. LEHD staff combine these with a range of other data sources to generate QWI, OnTheMap, and other new products under development. By integrating data used to administer public programs with existing census and surveys, a new national jobs database is generated at very low cost and with no additional respondent burden.[3]

The purpose of this paper is to describe the methodology by which national firm age and size data were integrated from other data sources (chiefly the Business Dynamics Statistics) into the LEHD data. In the first section we describe the input data sets and the approach for linking the different business data. The second section of this paper describes the core firm age and size integrated file, and our edits and imputation methodology. In the last section we benchmark the distribution of firm age and size in the QWI to other published data sources and provide a short example of how the data can be used, relying on the first release of Maryland's firm age and size data.

B. Input Files

The BDS microdata file

To add firm age and firm size to the LEHD data we rely primarily on data from the Business Dynamics Statistics (BDS) program.[4] The BDS microdata contain annual longitudinal firm-level

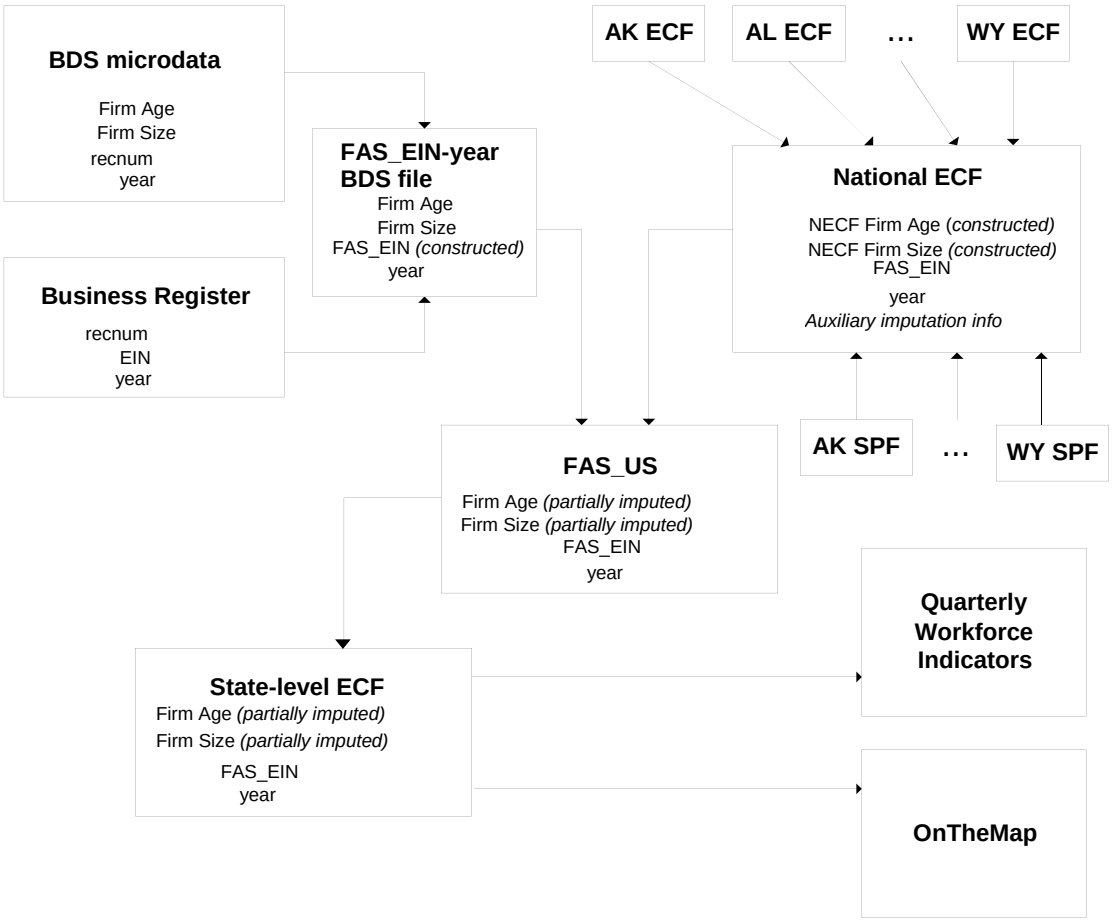
[2] Data for Massachusetts, Puerto Rico, and the US Virgin Islands are under development.

[3] For a complete description of the LEHD data see Abowd et al (2009).

[4] The BDS are tabulated from an enriched version of the Center for Economic Studies' Longitudinal Business Database (LBD). The BDS program staff provides to LEHD an intermediate microdata file which contains the BDS estimates of age and size - we refer to it throughout this document as the BDS microdata file. For more detailed descriptions of the LBD and BDS, see Jarmin and Miranda (2002) and Haltiwanger, Jarmin and Miranda (2009).

information for nearly the entire non-farm private economy and a small portion of the public sector from 1977 through the present. A unique feature of this data are the longitudinal identification and analysis of firms through ownership and structural changes such as mergers, thus yielding more accurate estimates of firm age and firm size. Rather than reconstruct such longitudinal firm linkages in the LEHD business data, our approach is to use common firm identifiers to link between the two frames and use BDS-sourced estimates of size and age wherever possible.

Figure 1: A Simplified Data Flow Diagram for the Inclusion of Firm Age and Size into the LEHD Infrastructure Files, the QWI and OnTheMap



The BDS microdata rely on the Business Register, the Census Bureau frame for firm and establishment surveys, to create firm-level longitudinal links. Specifically, the Business Register has data at three levels of business activity: 1) the business establishment (a physical location where economic activity occurs), 2) the Employer Identification Number (EINs) (the tax-filing

unit), and 3) the enterprise (the parent firm that controls more than 50% interest in the related establishments). When creating the BDS, these identifiers from the Business Register are further augmented by probabilistic establishment name and address matching. The result is the "alpha" level identifier, a firm identifier that is sensitive to changes in ownership and firm structure. Longitudinally, the alpha firm may be composed of different EINs and different establishments as firms are acquired or merge, for example.

National firm age is defined as the age of the oldest establishment when a new alpha comes into existence. A new establishment of an older firm is assigned the age of the national firm. The firm then ages naturally regardless of the composition of establishments at the firm. Establishments, however, *can* exhibit abrupt changes in firm age if, perhaps during a merger or acquisition event, the parent firm of the establishment changes. As described in greater detail below, the LEHD-BDS data integration is at the federal EIN level. So, in the LEHD data, establishments can exhibit abrupt changes in age when their EIN changes (as is typical when ownership of the establishment changes).

National firm size is the sum of all workers employed at each establishment in the national (alpha) firm on March 12th of the year. For several reasons, including the time lag between the BDS and LEHD data (the latter is several quarters more current), we use initial firm size. Initial firm size for any given consecutive two-year period is defined as the employment-weighted sum of firm size on March 12 in year *t-1* of all establishments that are part of an EIN on March 12 in year *t*. This definition automatically covers mergers, divestitures, acquisitions, etc. For instance, if a firm in year *t* has three establishments belonging to three different firms in year *t-1*, initial firm size in year *t* is the weighted sum[5] of the firm sizes in year *t-1* of each of these three establishments.

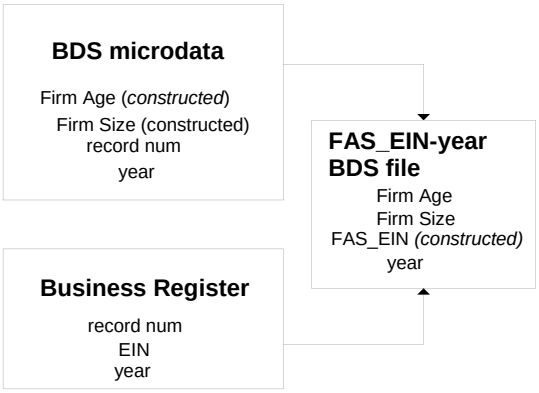
There exists no common establishment-level identifier between the BDS and LEHD data.[6] Instead, we merge the data via their shared common identifier, the federal employer tax identifer

[5] The weights are based on the year *t* size of each establishment.

[6] The U.S. federal statistical system has two business registers, the Census Business Register and the Bureau of Labor Statistics business register (the Quarterly Census of Employment and Wages, or QCEW). The BDS uses the Business Register frame while LEHD uses the QCEW frame. While there is substantial overlap between the two

(EIN). The first stage in the linkage is to merge the employer identification number (EIN) from the Business Register onto the BDS to create an input file for LEHD data integration. These steps are shown in Figure 2. Because the EIN is not complete in the LEHD data, however (it is missing approximately 4% of the time), the EIN itself cannot serve as the unique identifier for the linkage file. Instead, a separate identifier, the FAS_EIN (Firm-Age-Size EIN) is constructed, which preserves the link between the EINs (when available) in the two data files. More information on the construction of the FAS_EIN is provided in Appendix II, but it can be thought of as equivalent to an EIN when the EIN is not missing, and equivalent to a state tax identifier (SEIN) otherwise. Attaching the EIN and creating the FAS_EIN is the final step in the creation of the BDS input file, a EIN-year level file, with firm age and size both calculated at the alpha firm-year level.

Figure 2: Creating the BDS Integration File



There are some notable differences in the scope of the BDS and LEHD data that should be kept in mind. The BDS data include nearly all non-farm private establishments with paid employees, as well as a limited set of public sector activities. The LEHD data, on the other hand, include all employment that is covered by the state unemployment insurance (UI) systems, including state government employees.[7] Initial tests confirm that NAICS sectors 11 (Agriculture, Forestry,

registers, business list comparison projects have noted many discrepancies between the two frames. These disagreements combined with a lack of common establishment identifier would make an establishment-to-establishment match an extremely ambitious undertaking.

[7] The areas not covered by UI include some agricultural employment, independent contracting, self-employment, military employment, federal civilian employment, railroad employment, some elected state and local

Fishing and Hunting) and 92 (Public Administration) appear to be better represented in the LEHD data than in the BDS data. In order to limit the impact of this data limitation on the firm age and size estimates in the QWI and OnTheMap, firm size and age are being released only for private sector jobs.

The National Employer Characteristics File (NECF)

LEHD-based estimates of firm age and firm size are used to edit or impute missing firm age and size data. The National Employer Characteristics File (NECF) serves as the source of LEHD-based information. As seen in Figure 1, the NECF is generated by interleaving all the state Employer Characteristic Files (ECF) created in the previous quarter's QWI data processing cycle. The ECF's chief data input is the Quarterly Census of Employment and Wages (QCEW), which reports quarterly employment and wages for virtually all private sector employment covered by the UI programs. These are augmented with the LEHD's Successor-Predecessor (SPF) state files, which distinguish between spurious changes of the firm identifiers and merger/acquisition events in the LEHD data.[8] The NECF contains three important pieces of information used in the creation of the integrated BDS/LEHD FAS_US file: the FAS_EIN (the firm identifier used to link to the BDS microdata), NECF firm size, and the year and quarter in which each firm first reports a positive payroll in the ECFs.

Firm age and size are estimated from the LEHD data so as to be as consistent with the BDS as possible. NECF-based firm age is the number of years that have elapsed since the FAS_EIN is first found in the NECF with a positive payroll. As in the BDS, firm age in the NECF is longitudinally consistent since a firm ages continuously from its first instance of positive payroll. These estimates of firm age suffer from the geographic and time series limitations of the LEHD data, which is a much shorter time series than the BDS. If a firm is born in a state that did not provide data for the year of birth, firm age will be underestimated. Since the earliest LEHD states begin in the 1990's, with many states continuing to enter the data universe well into the next

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government officials and postsecondary work-study students (Stevens, 2002). Data from the Office of Personnel Management are currently in development to be used for federal civilian employment.

[8] See McKinney and Vilhuber (2005) for a detailed description of the ECF. See Abowd et al. (2009) for a thorough overview of the SPF.

decade, many firms are left-censored in the LEHD data. These firms have their birth year imputed if no BDS match can be found.[9]

NECF-based firm size is the sum of monthly employment for each establishment associated with the FAS_EIN in March of that year. For firms born after the first quarter of a given year, the firm size is based on either the first month of positive employment in the quarter of birth. As in the BDS data, initial firm size is equal to March employment in year *t-1* while, for firms born in year *t*, it is equal to firm size at birth. Unlike the BDS, which uses period *t-1* employment of establishments belonging to the FAS_EIN in period *t*, firm size in the NECF is not based on establishments. Instead, it is the employment level associated with the FAS_EIN in year *t-1*. Hence, NECF firm size measures do not adjust for compositional changes in the firm, such as acquisitions or mergers.

Inspection of a matched BDS-NECF set of firms suggests that the national size of the EIN in the national ECF is highly correlated with the BDS alpha firm size (about 0.8) and, in about 80% of cases, firms in the matched data fall in the same size class tabulation category. Since the NECF data are not truly national, when the tabulation categories disagree, the NECF national firm size is almost always smaller.

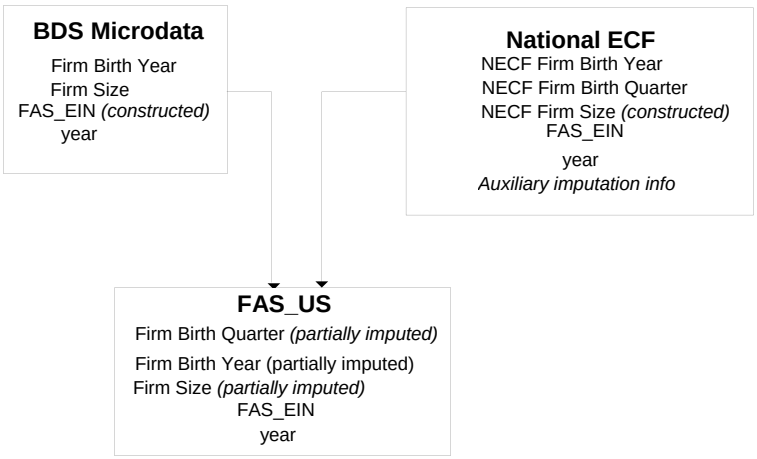
C. The FAS_US file

The Firm Age and Size national file, the FAS_US, is created prior to each quarter's production run to provide national firm age and firm size information as inputs into the quarterly LEHD infrastructure files. Figure 3 illustrates the creation of the FAS_US: it is created by merging the NECF and the BDS microdata by FAS_EIN and year, applying edits and imputes to complete the BDS-sourced measures of firm age and size. This section describes the data considerations that

[9] During the creation of the NECF firm age, a flag is generated to identify firms whose initial appearance in the NECF occurs in quarter q_o, where q_o denotes the first quarter of data available for a given state. For example, if a firm first appears in the data simultaneously in two states in a quarter that is q_o for at least one of these two states, the firm is flagged as being left-censored. This is because the firm may have existed in previous quarters in the left-censored state. Additionally, if a firm is associated with a predecessor-successor event, as identified via the SPF, in the year of its birth or the year after its birth, it is also flagged as being left-censored. The SPF identifies false firm births that are simply due to spurious changes in firm identifiers. These flags are later used as inputs into the imputation model for firm age.

motivated the definitions and sourcing of the firm age and size variables. It details how the two primary input data sources are combined and how each of the two new variables is calculated. Finally, it presents some results from initial quality assurance tests and summary statistics of the FAS_US file.

Figure 3: National Processing Data Flow Overview



Timing Differences of the LEHD and the BDS

The enhancement to add firm age and size to the LEHD production process was designed to ensure that firm age and size tabulations included as part of the QWI quarterly data release are primarily based on BDS microdata. Since the BDS is based on tax filings, it lags the availability of most LEHD files by several quarters. Two decisions were made to handle the significant lag in the availability of BDS data relative to other LEHD files: 1) firm size is based on a one-year lag when possible, and 2) the most recent quarter of firm age and size data are not released.

As shown in Table 4, the timing of the production cycle of the Quarterly Workforce Indicators for new and re-submitted quarters differs substantially from the timing of the BDS microdata. Each quarter, Local Employment Dynamics state partners submit data for the third quarter prior and resubmit data for the fourth quarter preceding the quarter during which production is taking place.[10] For example, in the production run that occurred in the first quarter of 2012, data for the

[10] The latter of these quarters of data is a re-submission that contains a typically small number of corrections or late-reported records.

second quarter of 2011 was submitted for the first time, along with a resubmission of data for the first quarter of 2011. On the other hand, since the BDS microdata are based on business tax data, it is not available for a given year until November or December of the following calendar year. For example, 2010 BDS microdata was not available until November of 2011.

Table 4: The Data Inputs of the First Seven Quarters of Firm Age and Size

Production Cycle	Most Recent State UI/ES202 Submission	State UI/ES202 Re-submission	Most Recent NECF Information	Most Recent BDS Submission Date	Year of Most Recent BDS Information
2011Q4 *Oct-Dec 2011*	2011Q1	2010Q4	2010Q4	October 2010	2009
2012Q1 *Jan-Mar 2012*	2011Q2	2011Q1	2011Q1	November 2011	2010
2012Q2 *Apr-Jun 2012*	2011Q3	2011Q2	2011Q2	November 2011	2010
2012Q3 *Jul-Sep 2012*	2011Q4	2011Q3	2011Q3	November 2011	2010
2012Q4 *Oct-Dec 2012*	2012Q1	2011Q4	2011Q4	November 2011	2010
2013Q1 *Jan-Mar 2013*	2012Q2	2012Q1	2012Q1	November 2012	2011
2013Q2 *Apr-Jun 2013*	2012Q3	2012Q2	2012Q2	November 2012	2011

Since the BDS microdata offers a more accurate measure of national firm age and size than the current LEHD data, as previously detailed, and it is always at least one year older than the state-provided data, firm size in QWI and OnTheMap and infrastructure files is *lagged* firm size, referred to as initial firm size.[11] Taking advantage of the fact that the BDS microdata include lagged and current firm size, an additional year of BDS data is generated by extrapolating from the most recent year of data included. In this additional year of BDS data, age is generated by

[11] Lagged values from the BDS microdata are used to define size categories except for new businesses (those with previous size of zero) for whom firm size is their initial, or year of birth, size.

allowing firms in the most recent year of BDS data to age one year. Firm size is generated by setting lagged firm size in the new year of data equal to the previous year's current firm size. Since the firm universe is defined by the NECF, there are no downstream implications if firms that died in year *t-1* are included in year *t* in this step, since they will be dropped if they are no longer in the NECF. The only differences between the additional year of BDS generated data and the original input is that current size is always missing in the generated data and the lagged firm size that is created for the most recent year does not adjust for changes in firm composition.

Recall, further, that the BDS firm size measurement is based on the March 12th employment level of that year. The timing difference between the BDS microdata and the NECF shown in Table 4 means that a given year's fourth quarter production run, which creates the first quarter data release for that year, could either have its lagged firm size based entirely on the NECF or based on BDS firm size of March 12th of year *t-2*. Rather than one of these options, our solution is to not release the most recent quarter of firm age and firm size – hence, firm age and size in the QWI will be released one quarter after the initial QWI release for a given quarter. In this way, when the first quarter firm age and size data are released for a given year, they will rely on the most recently received BDS data, which is based on March 12th of year *t-1*. This decision has the additional benefit of ensuring that firm age and size always use NECF data that already incorporate any data resubmits that may have occurred during the previous production cycle, thus minimizing the number of new EINs introduced into the ECF process during the state-level firm age and size processing described later in this documentation.

In the unlikely situation in which BDS microdata is unavailable for a given year, the NECF will be updated and used in combination with older BDS microdata to create a new FAS_US. In this situation, NECF firm size data will supplant BDS firm size data. Firm age will be generated by allowing the most recent firm age available in the BDS to age naturally for continuing firms and assigning NECF-based age for new firms.

Table 5. Distribution of Firm Age and Size Source

		FAS_EINs		Employment	
Firm Age Source		%	Cumulative	%	Cumulative
BDS	BDS (year t)	81.4	81.4	94.6	94.6
	BDS (year t+1)	0.4	81.8	0.3	94.9
	BDS (year t-1)	4.1	85.9	0.8	95.7
	BDS (year t+2)	0.1	86.0	0.1	95.8
	BDS (year t-2)	1.9	87.9	0.3	96.1
NECF	NECF Earlier Birth	1.3	89.2	0.6	96.7
	NECF 0 Empl: Death edit	1.1	90.3	0.0	96.7
	NECF 0 Empl: Birth edit	0.5	90.8	0.0	96.7
	NECF Age	6.1	96.9	1.6	98.3
	NECF Edit of Impute	0.1	97.0	0.1	98.4
Impute	Impute	3.2	100.2	1.6	100.0
Firm Size Source		%	Cumulative	%	Cumulative
BDS	BDS - preferred size	69.4	69.4	92.8	92.8
	Lagged size from *t+1*	0.1	69.5	0.2	93.0
	Size from *t-1*	3.2	72.7	1.1	94.1
	BDS - other size (*t)*	5.4	78.1	1.7	95.8
NECF	NECF 0 Employment	5.2	83.3	0.0	95.8
	Impute, NECF reset	1.1	84.4	0.0	95.8
Impute	Impute	5.7	90.1	0.1	95.9
	NECF growth from impute	9.9	100.0	4.0	99.9

Source: Data from the 2013Q3 production run.

The Creation of Firm Age and Size

The FAS_US file contains completed annual firm age (firm birth year and quarter)[12] and firm size information for all FAS_EINs in the NECF file (the universe of firms in the LEHD data). Specifically, it contains NECF-sourced firm birth year and quarter, BDS-sourced firm birth year defined using payroll information (when available), completed firm birth year and firm birth quarter (firm_birth_year and firm_birth_quarter), lagged and current year's March employment based on the NECF data, lagged and current year's March employment based on the BDS data (when available), and best initial size ("firmsize"). Best initial size is lagged firm size for continuing firms or the current year's March employment for new firms. Firm_birth_year,

[12] Since the FAS_US file reports annual observations, firmage for the QWI and LEHD infrastructure files is calculated using quarterly data during the production process so as to smooth ageing patterns throughout the year.

firm_birth_quarter, and firmsize are the completed and edited variables used in the LEHD public data releases. This section outlines the edits and imputations that are applied to clean and complete these variables prior to production processing. Table 5 details the proportion of the FAS_US records in the 2013Q2 run by source of birth year and size information.

BDS-based edits

In order to maximize the match rate between the NECF and the BDS, our first edit is to allow for a two-year lag in the matching algorithm. If firm birth year is missing in year *t*, BDS records from years *t-2* to *t+2* are searched for non-missing payroll birth year. The nearest record with a valid birth year within the four-year window is used to edit the missing firm birth year observation.[13] This edit adjusts the matching window to accommodate for differences in birth/death/merger timing resulting from the fact that the LEHD data are quarterly, while the BDS are annual. For example, recall that firms are included in the BDS based on their March 12th payroll. New firms born after March 12th of that year will appear in the LEHD data a year before appearing in the BDS. Similarly, firms that were alive at the beginning of the year but died before March 12th may also be missing from the BDS but be present in the LEHD data that year. Experimenting with widening the search window showed that the match rate could be increased even more by widening it to two years in either direction. We assume this is because of genuine timing differences between the two data sources. In the 2013Q3 production run, this edit was used in the firm age of 6.5% of all FAS_EIN observations.

This same edit cannot be applied to firm size since changes in firm size between years is not deterministic to the same degree as age. Thus, the BDS-based edit for firm size is far more conservative: we restrict the edits to firm size to data from *t-1* and *t+1*. Specifically, we set initial firm size in year *t*, which should be firm size in March of the previous year, to firmsize in year *t-1*. When the firm matches to year *t+1*, firm size in year *t* is set equal to the lagged size in year *t+1*. These two edits only provide firm size for about 3.3% of all records. Finally, for firms with BDS data in year *t* missing the preferred firm size, we used the other BDS firm size (current year

[13] Only birth years from the BDS data that do not contradict NECF age are kept using this edit. For example, firms born earlier in the NECF than in the BDS do not rely on this edit.

instead of previous year's, for example) if it was available since the correlation between this year's and last year's firm size is typically very high.

Overall, the vast majority of firm age and size data are derived directly from the BDS. At the FAS_EIN level, over 81% of firm age data and 69% of firm size data come from direct matches to the BDS preferred variables. These FAS_EIN observations correspond to 95 and 93% of all jobs.

NECF-based edits

In a small share of cases a firm appears in the LEHD data before it is 'born' in the BDS. In these cases we generally favor the LEHD-sourced age of the firm, as the alternative is to have firms with negative ages in the public use data. However, we are careful to only use this edit if there is clear evidence that the firm is in fact active, and not making payments to UI accounts that are not associated with current employment. Three edits are made to firm age using this information:

1. In cases where a firm is born in the LEHD data up to 2 years earlier than in the BDS, where the firm reports both positive payroll and positive employment, firm birth year from the NECF is used to update all records until a change in BDS-based birth year is identified. Since a change in birth year probably indicates a change in ownership, we allow the firm birth year to revert to the BDS value.
2. In cases where the firm has a false birth in the LEHD data, determined by no positive employment in that calendar year despite positive payroll until after it is born in the BDS, then firm birth year is set to -1, indicating that the firm is inactive, until it is born in the BDS (inactive firms are by definition not tabulated in QWI and OnTheMap).
3. In cases where the firm dies in the BDS but continues to exist in the NECF with 0 employment, then the firm retains the same birth year even after it stops matching to the BDS. If, on the other hand, there is positive employment after death in the BDS, it retains the same birth year for the first two years, and then has birth year imputed for the remainder of the years (since a change in ownership may explain why it no longer matches to the BDS).

LEHD data are also used to edit firm size, but these edits are limited to two cases. The first is when the BDS reports 0 employment and the NECF does not show any positive employment in that year. In these situations, the 0 reported in the BDS size is taken as a true 0[14] and firm size is not imputed. The other case is when the NECF-based growth of an imputed size (described below) leads to 0 employment though the NECF reports positive employment. In these cases, the firm size is reset to the NECF size and allowed to grow from there.

Beginning with the 2013Q3 production run, we also produce data on quarter of birth. Since the BDS is an annual data file, our primary BDS-based firm age data input is limited to year of birth. However, since the LEHD data sets are quarterly, relying solely on year of birth means that all firms age one year in the first quarter of each year. This results in drastic changes in the age distribution at the beginning of each year. To address this, we impute quarter of birth; this allows the ageing distribution to be smooth over the calendar year. Birth quarter is sourced from the NECF for firms with valid births in the NECF. For left-censored firms, birth quarter is imputed based on the distribution of birth quarter within state and industry cells among the subset of firms that are not left-censored. This impute controls for differences in seasonal patterns of work and firm creation throughout different regions of the country and industries. Importantly, the primary purpose of this variable is to smooth the longitudinal analysis of the published quarterly data; it should *not* be used to research topics such as seasonal patterns of business creation.

Imputation

After all edits are exhausted, the remainder of the missing birth year and firm size values are imputed. The imputation strategy applied beginning with the 2013Q2 production run "imputes" missing birth year for firms that are not left-censored or associated with a successor-predecessor event by using the NECF-based birth year. As shown in Appendix III, NECF-based age was found to be a very accurate predictor of BDS age for firms whose births are captured in the NECF. For firms that are left-censored or associated with successor-predecessor events, year of birth is imputed only for the first year in which age is missing and reused for all subsequent years in which age is missing. Similarly, only firms missing firm size in their first appearance in the NECF have size imputed. For any subsequent observations missing size, firm size is determined

[14] A 0 in the BDS size data can represent either unknown employment (i.e., missing data) or no employment.

by allowing size to grow from the previous year at the same rate as it grew in the NECF. For example, if firm size in the NECF grew by 2% from year *t* to *t-1*, firm size from year *t-1* is increased by 2% and set as the value of firm size in year *t*.[15] In a sense, this strategy allows us to impute the magnitude, while using the NECF data as a growth edit. This methodology generates correlated size over the life of a firm, a quality not reliably reproduced using independent imputations for each year of missing size data. All imputed values are generated from a multi-step Bayesian imputation model relying on probability distributions from stratified non-missing observations as detailed in Appendix III.[16]

FAS_US File

Though the FAS_US file contains completed information on firm birth year and size for all FAS_EINs in the NECF, LEHD will release firm age and firm size tabulations only for privately owned firms. Limiting tabulations to privately owned firms will mitigate the effects of the differences in scope between BDS and LEHD data. This, in turn, limits our reliance on the NECF and imputations for firm size and firm age. Table 6 lists the variables found in the FAS_US file.

D. The New QWI Tabulations

This section details new QWI tabulations that incorporate the newly integrated information on firm age and size in the LEHD data. Table 7 provides a summary of the new firm age and size QWI tabulations. Firm age and firm size are calculated for NAICS sector and NAICS 3 and 4-digit groups at the state level, and by NAICS sector at county, Core Based Statistical Area (CBSA) and Workforce Investment Area (WIA) levels. All firm age and firm size tabulations are also available within demographic groups (sex x age, race x ethnicity, and sex x education). Note that tabulations interacting firm age and size are not released, as size and age are heavily correlated, and many cells of the off-diagonal would be thinly populated.

[15] Some edits and restrictions are applied to the growth rate when the previous year's initial size differs too greatly from the NECF size. For example, if a firm's NECF size grew from 1 to 2, this implies a 100% increase in employment. However, if the matching BDS size for the initial year is 100, then a growth rate of 100% is not in line with an actual increase of 1 job. In these situations, a more conservative growth algorithm is used.

[16] Prior to the 2012Q4 production run, firm size was set to the NECF lagged size for all observations that did not match to the BDS. The 2012Q4 and 2013Q1 production runs imputed all observations of missing size and age. These approaches are both detailed in Appendix III.

Table 6. Specifications of the FAS_US File

Variable	Type	Length	Label
ein_first_year	Num	3	First year positive payroll for EIN in NECF
fas_ein	Char	14	Completed EIN used for FAS, generated from SEIN if EIN is missing or invalid
firm_birth_quarter	Num	3	Completed quarter of birth
firm_birth_year	Num	3	Completed year of birth
firmage_pay	Num	8	LBD age: year-min(f_firstyear,intialyr)
Firmid	Char	10	Firm Alpha from LBD
firmsize	Num	8	Best initial firm size (size March 12 of last year, current size if new)
firmsize_t	Num	8	LBD national firm (alpha) March 12 employment size
firmsize_tm1	Num	8	LBD national initial firm size (March 12 employment in last year)
lbd_dob	Num	8	LBD birth year (year-firmage_pay)
lbd_match	Num	8	=1 if EIN matches to LBD this year, 2=next year, 3=previous year, 4=NECF(no LBD), 5=ECF
Mu	Num	8	LBD Single-Multi Identifier
necf_age_left_censored	Char	1	1 if NECF EIN 'birth' is the first quarter of data available
necf_ein_size	Num	8	ECF EIN size of firm on March 12 of this year
necf_ein_size_tm1	Num	8	ECF EIN size on March 12 of last year (current size if EIN birth year)
source_age	Char	1	1=LBD, 2=ECF, 3=imputed, 4=ECF
source_age_detailed	Num	8	detailed age source flag (NECF/LBD only)
source_size	Char	1	1=LBD, 2=ECF, 3=imputed, 4=ECF
source_size_detailed	Num	8	detailed size source flag (NECF/LBD only)
Year	Num	3	Year

Table 7. Firm Age or Firm Size Tabulations

Employee Universe	All statutory employees, ages 14-99 Sex x Age Race x ethnicity Sex x Education	
Ownership	Private sector only (ownership code = 5)	
Firm Age or Size Categories	All firm ages 0-1 years 2-3 years 4-5 years 6-10 years 11+ years	All firm sizes 0-19 employees 20-49 employees 50-249 employees 250-499 employees 500+ employees
Geography Detail	State County Metro (CBSA) WIA	
Industry Detail	NAICS sector 3-digit NAICS *(for state-level only)* 4-digit NAICS *(for state-level only)*	

Benchmarking the QWI Firm Age and Size Statistics

In this section, we benchmark the new QWI firm age and size employment data to two comparable data sources: the Census Bureau's Business Dynamics Statistics (BDS), and the Business Employment Dynamics (BED) produced by the Bureau of Labor Statistics. To allow for a longer time series, we limit the QWI analysis to twenty-eight states for which data are available from 1998.[17] Due to differences in scope and timing between the data sources, employment counts are not comparable.[18] Instead, we verify that the employment trends match. The following figures show that, overall, the QWI measures of firm age and size line up well with these other sources.

[17] The included states are Colorado, Connecticut, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Kansas, Maine, Maryland, Minnesota, Missouri, Montana, Nevada, New Jersey, New Mexico, North Carolina, North Dakota, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Washington, West Virginia and Wisconsin. All data used in this analysis is from the 2012Q1 production run.

[18] Employment is measured differently by the three data sets: it is March 12th employment in the BDM, average first quarter employment in the BDS, and April 1st employment in the QWI (beginning of period employment in quarter two). See Davis, Faberman and Haltiwanger (2006) for a comparison of job and worker flows between different data sources and a more detailed explanation of how these may differ.

Figure 4. Employment as a Proportion of 1999 Employment in the BDM, BDS, and 28 State Sample of the QWI, 1999 - 2010

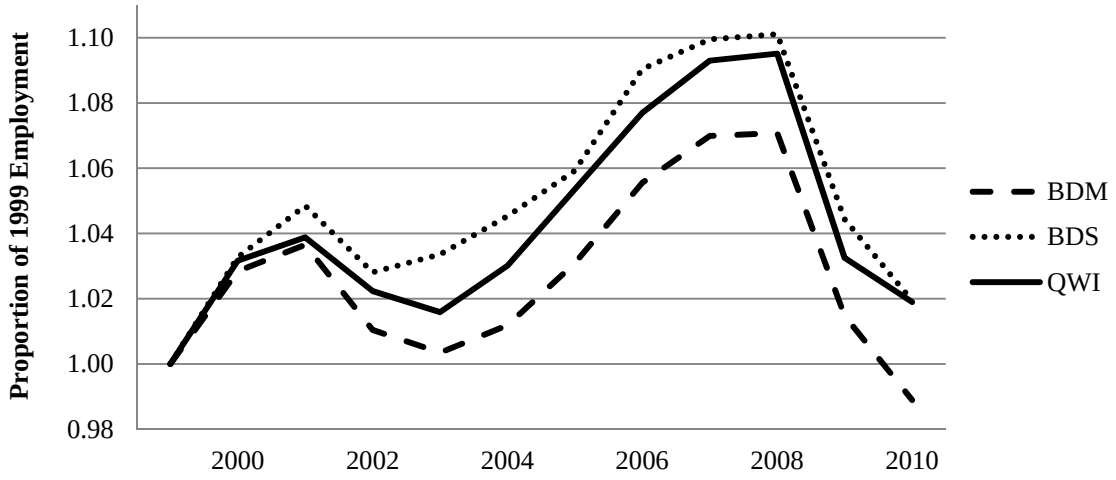


Figure 4 compares the employment trends reported in the QWI to those reported in the BDM and BDS published statistics. The series are reported as a proportion of 1999 employment, allowing for more direct trend comparisons between the three data sources. Figure 4 illustrates that the QWI capture the cyclical employment changes of the past decade in line with the BDM and the BDS.

Figure 5. Employment in Small and Young Firms, as a Share of Total, 1999 - 2010

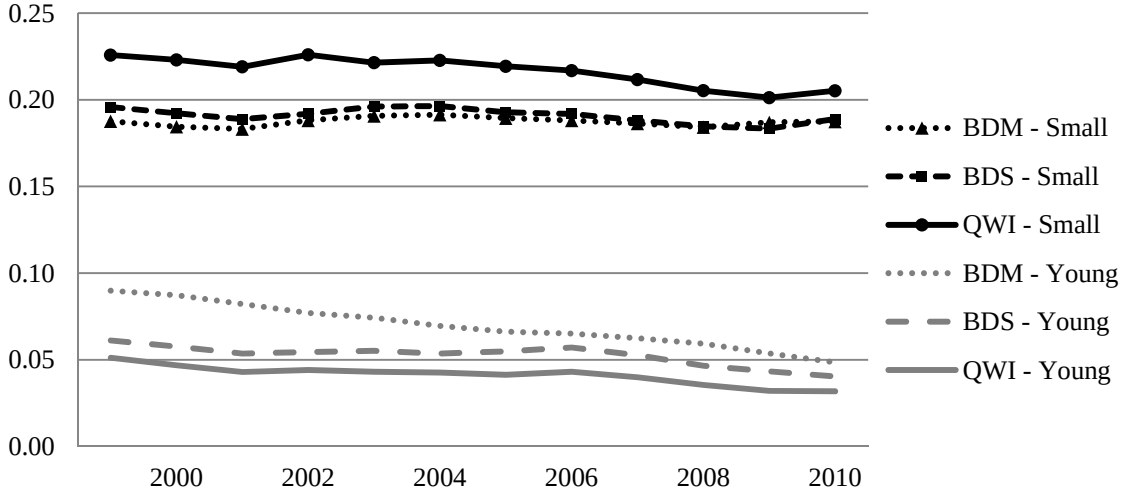


Figure 5 illustrates the share of total employment attributable to either small or young firms in each of the three data sources. Note that the BDM young firm statistics are calculated at the *establishment* level, rather than the firm level: when an older firm opens a new establishment, the employment from that establishment is attributed to the young category in the BDM series, but to the older category in the BDS and the QWI.

Figure 5 shows that the QWI include more employment in small firms and fewer jobs in young firms than the other two data sources. Part of this difference may be due to the twenty-eight state sample used for the QWI: this sample of states may not be perfectly representative in the firm age and size distributions. Even with these differences, the similarity in trends for both small firms and young firms is encouraging. As in the other data sources, the QWI show a fall in the number of small firms beginning in mid-2006. Similarly, while the BDM show a decline in the number of young establishments from 1998 onwards and the BDS show some increase in young firms around 2006, the QWI trend shows elements of both - a slight decline in the early 2000s, followed by a steeper decline after 2006.

Next, we look at the trends in job creation rates by firm size using the BDM and the QWI, the only two sources for which this comparison is possible.[19] The net job creation rate of firm *i* in period *t* is calculated as follows:

$$Net_JC_{it} = \frac{JC_{it} - JD_{it}}{Avg(E_{it})} = \frac{E_{it} - E_{it-1}}{\frac{1}{2}(E_{it} + E_{it-1})}$$

where E_{it} designates the average employment of firm *i* over period *t*. The net job creation rate is the difference in the total number of jobs at two points in time as a proportion of the average number of jobs over these two periods. It can be equivalently calculated using job creation and job destruction totals or using employment totals in the two time periods, as shown above. This formula is applied to the BDM and QWI differently to accommodate the different data available

[19] This comparison cannot be done for firm age since the BDM reports employment by establishment age and the BDS only provides job creation and job destruction rates annually. Similar annualized measures of net job creation cannot be generated from the QWI tabulations since we cannot control for firm-specific age or size changes within the year.

in each. While the denominator for the BDM series is the average employment level over the first quarter of each year, the QWI series uses the average of beginning (B) and end (E) employment level in the second quarter of each year as the denominator. Similarly, JC and JD are calculated over quarter two for the QWI but as the change in employment between December 12th and March 12th (the third month of the fourth and first quarters) in the BDM.

Figures 6 and 7 show the net job creation rates for firms that have 20 or more employees and for those with less than 20 employees. Once again, though the levels are not exactly the same, the overall trends are consistent between the QWI and the BDM. Both show a dip in 2001, some recovery in the middle, and a severe dip in 2009. It should be kept in mind that there is a significant methodological difference in how job creation and destruction are measured in the

Figure 6. Comparing Net Job Creation Rates in the QWI and the BDM, for Firms with 20 or More Employees, 1999 - 2010

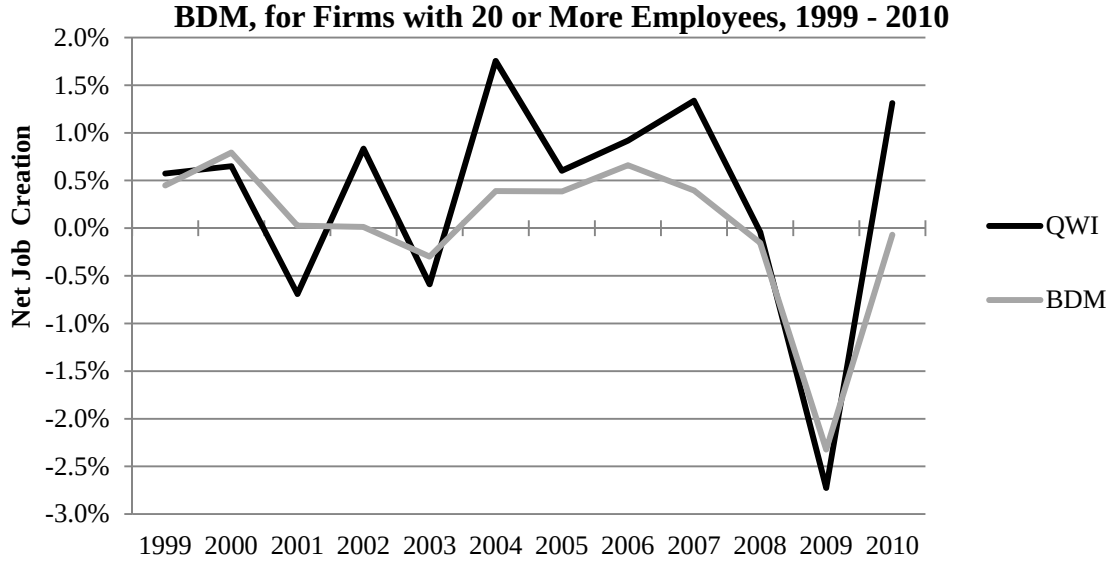
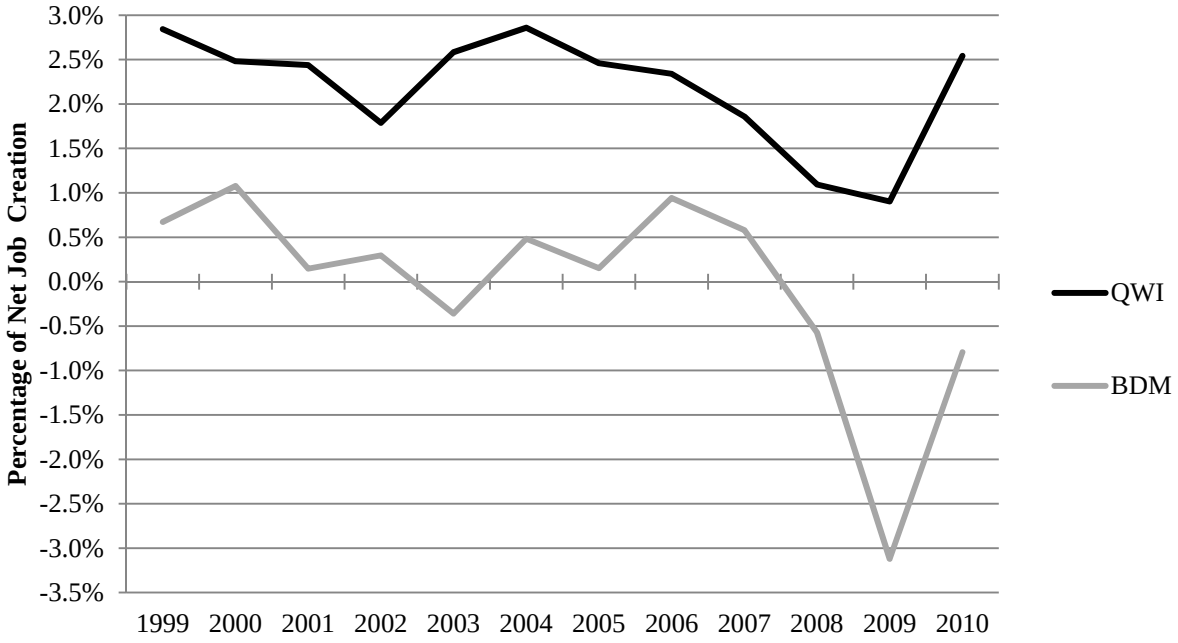


Figure 7. Comparing Net Job Creation Rates in the QWI and the BDM, for Firms with 19 or Fewer Employees, 1999 - 2010



BDM and the QWI. This methodological difference explains much of the difference in net job creation rates for smaller firms. The BDM uses the concept of *dynamic sizing* in attributing employment changes by firm size while the QWI use initial firm size[20]. Dynamic sizing attributes job creation and destruction to the current size of the firm. For example, if a firm that began with 10 jobs was to create 20 new jobs, the QWI would attribute all of these new jobs to the size category of 19 or less employees. On the other hand, the BDM would attribute the first 9 jobs to the smaller firm size and the remaining jobs to the 20+ size category. As a result, dynamic sizing results in systematically lower job creation rates for smaller firm size groups. For larger firms, the two methodologies yield very similar results since larger firms are less likely to fluctuate between firm size groups.

Beta QWI Statistics for Maryland

To illustrate the firm size and age tabulations, we have included a sample of QWI statistics calculated for the first quarter of 2011 for the state of Maryland. Table 8 lists different employment count variables and job creation/destruction variables that are standard products of

[20] For a thorough comparison of these two methodologies and their implications for job growth measurement, see Okolie (2004).

Table 8. Sample QWI Statistics for Maryland, All Industries, 2011Q1

Firm Age	Beginning of period employment (B)	End of period employment (E)	Accessions (A)	Separations (S)	New Hires (H)	Job Creation (JC)	Job Destruction (JD)
0-1 years	44,414	54,766	24,418	14,066	23,280	14,678	4,339
2-3 years	68,196	69,799	16,208	14,605	13,580	6,418	4,828
4-5 years	67,194	68,701	13,927	12,420	11,842	5,756	4,262
6-10 years	175,823	177,811	30,056	28,068	25,452	11,799	9,802
11+ years	1,570,760	1,559,112	173,094	184,742	148,521	52,073	63,592
Firm Size							
0-19	359,029	371,396	68,036	55,669	56,887	35,999	23,833
20-49	186,839	188,616	27,721	25,943	23,419	10,593	8,825
50-249	322,392	323,211	43,056	42,237	37,479	12,810	12,009
250-499	107,011	107,418	13,684	13,277	11,981	4,038	3,687
500+	936,585	924,984	102,992	114,593	91,058	26,285	37,669
Total	1,926,402	1,930,317	257,858	253,943	222,829	90,755	86,811

Source: Data from the 2012Q1 production run. Note that this beta release relied on a simple firm age impute only for left-censored or successor-predecessor firms. All other missing firm age and size observations were edited with the NECF-based measures.

the QWI, by firm size and by firm age. An initial analysis reveals, for example, that all firm size and firm age groups gained jobs in the first quarter of 2011 in Maryland, except for the oldest and largest firms. While net job creation (JC-JD) indicates relatively large job gains in firms that are 0-1 years old and those who have 19 or less employees, the same measure shows that firms with 500+ employees and those that are 11 or more years old shed over 11,000 jobs. However, since these large/older firms account for the bulk of employment, these jobs represented less than 1% of all jobs among these firms. Though the QWI will not tabulate these values for firm age interacted with firm size, it is evident from this table that there is a significant overlap between large firms and those that are 11 or more years old.

The ability to consider the different employment dynamics for full quarter jobs versus all jobs will allow policy makers and researchers to begin to answer questions regarding *quality* of jobs being created by small firms versus large firms and by start-ups and young firms versus older firms. Table 9 reports some employment statistics for full quarter employees (those who were working at the firm throughout the entire quarter). Unlike the results for all jobs reported in

Table 9. Sample QWI Statistics for Full Quarter Jobs in Maryland, All Industries, 2011Q1

Firm Age	Employment (F)	Accessions (FA)	New Hires (H3)	Job Creation (FJC)	Job Destruction (FJD)
0-1 years	36,308	8,691	8,287	6,963	2,903
2-3 years	58,204	7,733	6,666	4,717	3,806
4-5 years	58,275	6,935	5,936	4,138	3,507
6-10 years	155,160	16,090	13,802	9,374	8,142
11+ years	1,426,338	109,869	97,410	48,458	43,694
Firm Size					
0-19	320,535	30,794	24,735	23,852	20,179
20-49	167,395	14,747	12,813	8,127	7,625
50-249	290,748	25,665	22,347	12,139	9,411
250-499	96,757	8,191	7,533	3,527	2,747
500+	845,916	68,676	63,676	25,132	21,394
Total	1,734,299	149,318	132,102	73,610	62,039

Source: Data from the 2012Q1 production run. Note that this beta release relied on a simple firm age impute only for left-censored or successor-predecessor firms. All other missing firm age and size observations were edited with the NECF-based measures.

Table 8, full quarter jobs increased across the board: all firm age and firm size groups reported positive full quarter net job creation.

Another important indicator of job quality is captured in the average earnings reported in Table 10. One striking fact that is evident at first glance is that small firms pay significantly less, even to full quarter employees, than large firms. Average monthly earnings for full quarter employees increase from a low of $3,082 in firms of less than 20 people to a high of over $5,000 for firms of 500 or more employees. Part of this striking difference in earnings may be due to a difference in the composition of the workforce at large and small firms in Maryland. For example, perhaps large firms are more likely to be found in high wage sectors than small firms. Since the QWI Online also reports earnings by industries/sectors, this question of firm composition is easy to explore. Indeed, a secondary analysis of the Maryland data by industry reveals that the highest paying sectors are also likely to be dominated by large firms: utilities, management of companies, finance, and professional/scientific services.

Table 10. Sample QWI Statistics of Average Earnings and Non-Employment in Maryland, All Industries, 2011Q1

Firm Age	Average Full Quarter Employment Earnings (Z_W3)	Average Change in Earnings for Accessions (Z_dWA)
0-1 years	$3,247	$488
2-3 years	$3,012	$610
4-5 years	$3,179	$631
6-10 years	$3,674	$808
11+ years	$4,504	$1,038
Firm Size		
0-19	$3,082	$747
20-49	$3,627	$673
50-249	$4,007	$713
250-499	$4,310	$717
500+	$5,025	$1,192
Total	$4,309	$909

Source: Data from the 2012Q1 production run. Note that this beta release relied on a simple firm age impute only for left-censored or successor-predecessor firms. All other missing firm age and size observations were edited with the NECF-based measures.

E. Conclusion

Beginning in 2012, the LEHD program has produced statistics on employment by firm age and firm size. Firm age and size are largely derived from the Census Bureau's BDS microdata, a reliable source of national firm-level information. Initial firm size is measured as the employment total in March 12th of the previous year while firm age is based on the age of the oldest establishment owned by the firm. These data follow the same format as existing released data and will employ the same confidentiality protection measures, along with additional noise added to the firm size tabulation variable.[21] These new tabulations, available both via the QWI Online and future releases of the OnTheMap tools, will allow policy makers and researchers to identify employment trends that often differ significantly by firm age and firm size. Recent topics of public discussion, such as the role of start-ups in creating new jobs, can be critically and accurately accessed at the local level, allowing for a better understanding of what type of firms are creating jobs and what type of jobs are being created.

[21] See Appendix V for details on the confidentially protection steps taken.

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Appendix I

Table of Acronyms

BDS: Business Dynamics Survey

ECF: Employer Characteristics File

EIN: Employer Identification Number

FAS: Firm Age and Size

FAS_EIN: Employer Identification Numbers developed for FAS project

LBD: Longitudinal Business Dynamics

LEHD: Longitudinal Employer-Household Dynamics

LODES: LEHD Origin-Destination Employment Statistics Dataset

QCEW: Quarterly Census of Employment and Wages

QWI: Quarterly Workforce Indicators

SEIN: State Employer Identification Number (State-level UI account identifier)

SEINUNIT: SEIN Establishment identifier

UI: Unemployment Insurance

Appendix II

Creation of the FAS_EIN

A new firm identifier, the FAS_EIN, was developed to link the BDS microdata and the NECF. Since the EINs are obtained directly from the Business Register, the creation of the FAS_EIN in the BDS microdata is straightforward: the EIN is simply pre-appended with five zeroes. The corresponding enhancement to the LEHD infrastructure files, however, must correct missing, partially missing, or invalid EINs. Specifically, three issues are addressed: basic typographical errors, SEINs with no reported EINs, and SEIN recycling. For SEIN records that contained no EIN in the ECFs, a pseudo-EIN is generated. Additionally, many states practice "SEIN recycling," that is, they reassign the state-level UI account numbers (known as SEINs) of dead firms to new firms. The FAS_EIN enhancement adjusts for SEIN recycling. This allows for the identification of new firms with recycled SEINs as separate entities from the dead firms that shared the same SEINs. The FAS_EIN was incorporated through modifications to the ECF production code during the first production cycle that added firm age and size to the infrastructure files, and is updated quarterly through the introduction of new firms.

Table A1: EINs by State

Description	Frequency
EIN not blank or invalid	95.3%
EIN blank	4.0%
EIN invalid	0.7%

Source: Data from the 2011Q1 production run.

EINs at the establishment level are provided by states in the QCEW dataset submission on a quarterly basis. Two major EIN-related data quality problems have been identified. The first is that EINs are sometimes missing or invalid (for example, the EIN is a string of 9's), while the second source of error is simply incorrect input. As shown in Table A1, slightly less than 5% of EINs are blank or otherwise invalid. This error frequently occurs during the first few quarters of an employer's existence. A substantial proportion of blank values can be filled through a simple longitudinal edit: when a SEIN appears with an EIN in some quarters but not others, the reported EIN is applied to all quarters.

However, it is not always appropriate to perform a longitudinal edit because of "SEIN recycling." A reasonable proxy for whether a SEIN has been recycled is that its EIN changed; Table A2 reports that 1.4% of SEINs have EINs that vary by spell. This phenomenon is especially pronounced in a subset of five states where SEIN recycling accounts for over 3% of submitted SEINs. Three of these have recycling in excess of 6% of all employers. Given these high rates of SEIN recycling in some states, it is especially important that longitudinal edits do not mistakenly apply the EIN from one firm to the missing values of another.

Table A2: SEINs with multiple EINs over time

One Spell	Multiple SEIN Spells	
	Single Dominant EIN	*Dominant EIN varies by spell*
91.1%	8.1%	1.4%

Source: Data from the 2011Q1 production run using SEINs from 48 states, years 1990-2010.

For national firm age and size integration, the EINs were processed as follows. First, any non-numeric characters were eliminated from the reported EIN and the field was left-filled with zeros to ensure that the EINs were nine characters long. This zero-padding most often changes EINs for which information underwent a character to numeric to character conversion that resulted in a leading zero being dropped.[22] The first two digits of the EINs are compared with a list of valid two-digit IRS Revenue District code, and any EINs that do not match a valid prefix are ignored.[23] An auxiliary file containing all EINs that ever appear in the BDS is used to identify any EIN that was ever reported in the BDS; these EINs are flagged, indicating they are the preferred EIN. Each sequence of reported SEIN information is broken into spells, where a spell ends when there is a four quarter or more gap in a SEIN's reporting. Otherwise, a new SEIN spell is identified if a SEIN switches its EIN consistently for four or more quarters.[24] For each SEIN spell, a dominant EIN is selected by calculating the modal EIN and the modal EIN that links to the BDS. If these match, then that is designated the dominant EIN. Otherwise, this implies that the most frequently occurring EIN in the LEHD data is not the one that links to the

[22] This type of error is generally found in an entire quarter of data for a state.

[23] The overwhelming majority of these are EINs that were initially submitted with seven numeric characters or less and zero filled: the prefix '00' is not valid. Note that this implies that any EIN that is corrected through zero-padding is only used as a FAS_EIN when only a single zero is prepended.

[24] Since states sometimes report an EIN initially with minor errors and subsequently report a corrected EIN, the spell identification method groups these slightly different EINs into a single spell. We correct for this by allowing SEIN spells to continue if the Levenshtein distance between the old and new EINs is less than 3.

BDS. In these cases, the EIN that links to the BDS is chosen if it accounts for more than 10% of the total frequency of the two modal EINs. Otherwise, the modal EIN that does not link to the BDS is selected as the dominant EIN. If all of these measures result in no EIN, a last attempt to find the EIN is undertaken by searching the UI-wage files sent by the states for a valid EIN. Finally, pseudo-EINs are assigned for spells that do not contain any EINs.

Table A3: Distribution of FAS_EIN flag

Flag Value	Description	Frequency
1	Listed EIN used	93.8%
2	SEIN-spell dominant EIN used, EIN missing or invalid	3.7%
3	Dominant EIN used, replacing valid EIN	0.5%
4	SEIN spell never had valid EIN listed, pseudo-EIN assigned	1.5%
5	No dominant EIN, EIN from UI is being used	0.6%

Source: Data from the 2012Q2 production run, averaged across SEINxSEINUNITxYEARxQUARTER observations.

Information about the FAS_EIN is contained in the variable fas_ein_flag. Table A3 contains the frequency, calculated at the establishment and quarter level, of FAS_EIN results from the NECF created for the 2012Q2 production run. In 93.8% of cases, the FAS_EIN uses the EIN (potentially with a left-padded zero) that is listed in the ES202. The majority of the remaining records (3.7% of all) had EIN missing and were filled in with another EIN from the same SEIN spell. Observations in which the EIN listed in the record were ignored and the dominant EIN was used instead accounted for only 0.5% of records. Another 1.5% of records had a pseudo-EIN assigned since they were part of a SEIN spell that never had an EIN reliably assigned. Finally, less than 1% of EINs were not found on the ES-202 but were found in the UI-wage files. These EINs do not undergo the same checks for spell dominance outlined above since they are only used as a last resort.

Work is currently underway to further improve the quality of the EINs in the NECF by incorporating a search in the Business Register and Geo-coded Address List (GAL) files. Preliminary evidence suggests that the Business Register may yield data on less than 1% of the observations that are currently not matching to the BDS. The GAL contains unique commercial and residential addresses geocoded to the Census block. This enhancement will allow firm identification based on business name and address, details available in the QCEW and the

Business Register. In this way, the GAL acts as a secondary linking mechanism between the SEINs in the QCEW and the EINs on the Business Register. All records in the NECF that do not match to the BDS will be candidates for this additional EIN search. This enhancement is planned for inclusion in data releases in late 2013.

Appendix III

Firm Age and Size Imputation Model

Once the data edits described in section C are completed, the imputation process is used to fill in any missing age or size observations. As is typical of real world applications, we begin by noting that the data missingness in this application is not random: in particular, small firms, young firms, and short-lived firms are less likely to match to the BDS. Though non-random missingness is not ideal, it is typical of data missingness in surveys and administrative data sources and many methods have been developed to address it. The imputation strategy utilized in this project has evolved over the lifetime of the FAS beta data releases: initial releases relied on simple NECF-based edits for size and age for most firms (Method 1), while the 2012Q4 and 2013Q1 releases relied solely on imputes (Method 2). Subsequent releases rely on a combination of the two previous models (Method 3).

Description of Method 1

The problem of data missingness in our particular application is partially mitigated by the high quality of the alternative firm age and size measurements available from the NECF. We know from preliminary analysis that BDS size and age correspond well to NECF size and age in matched BDS-NECF samples (see Tables A6 and A7 for an example of this comparison using a matched sample). An obvious substitute for missing BDS size and age is the NECF data; Method 1 does exactly this for all records missing size and for most that are missing age. For those firms that are left-censored or are associated with a successor-predecessor event, the NECF-based age is likely an underestimate of the true age. A simple hot-deck imputation model was developed to complete age for this subset of firms. Specifically, all BDS date of birth data was stratified by year and firm size category, and year of birth was assigned a probability corresponding to its frequency within each cell. Each FAS_EIN with missing firm age was similarly stratified and then assigned a year of birth using these probabilities. The year of birth was only imputed for the first missing observation and retained for all subsequent observations with missing age.

Description of Method 2

Rather than relying on NECF-based edits for size and most missing age observations, Method 2 imputes values for all missing observations of size and age. This imputation model relies on methodology similar to that used to impute race, ethnicity, gender, and age to the LEHD Infrastructure files. Specifically, the model relies primarily on a Bayesian imputation approach in which records without missing data are stratified by relevant explanatory variables and used to generate nested distributions of firm age or size (the posterior predictive distribution). These distributions are then used to generate the needed imputes. There are two significant differences between the hot-deck model used in Method 1 and the Bayesian model developed in Method 2: 1) more stratifiers are used in Method 2, and 2) Method 2 generates a distribution, thus allowing for the possibility of selecting a value that is not necessarily found in the data.

Method 2 addresses the issue of non-random missingness by maximizing homogeneity over the observable characteristics of the firms (NECF data) in each group in order to minimize any bias that springs from systematic, identifiable differences between included and excluded firms. For example, since we know from preliminary analysis that young firms are more likely to be smaller, we stratify by size when imputing age and we stratify by age when imputing size. In this way, when we impute age for a small firm, we are drawing from the distribution of age already controlling for being a small firm. As more stratifiers are added and the homogeneity of the subsample of firms increases, the expected bias in missingness falls.

The imputation model utilized here relies on stratifying and imputing separately for the firm age and firm size imputations. As in Method 1, firm birth is only imputed for the first record in which a firm is missing age and then reused for all subsequent observations missing age data. Firm size is imputed only when it is missing for the birth year, and then allowed to grow at the same rate as NECF-based size.[25] For each imputation group, we take the following steps (where "variable of interest" refers to the variable being imputed – either age or size):

[25] This strategy can be thought of as an NECF-based edit of firm growth in cases where it is missing from the BDS. The 2012Q4 and 2013Q1 production runs' beta releases of firm age and size relied on independent firm size imputations for each year in which firm size was missing. This strategy, however, failed to replicate desired firm size autocorrelation.

1. First, we choose a set of stratifiers; this is a set of never missing NECF variables that are related to the variable of interest.[26] These stratifier values are coded using nested "cell IDs", allowing for easy identification of each observation's stratification group. For example, if our stratifiers were NECF-size quintiles and NECF age quintiles (in that order), then this approach would yield 25 cell IDs: within each quintile of NECF-size, there would also be five nested quintiles of NECF- age.[27] The set of stratifiers is first generated using records without missing values of the variable of interest, the training data. Stratifiers fall into two broad groups: quantiles and fixed groupings. Quantiles are generated using continuous variables and their relative value to other observations within the same stratified group. Quantile cut-off points are stored in a look-up file since they typically vary between cell IDs. Fixed groupings stratifiers are assigned into cell IDs based on their exact values, hence no cut-off point is stored. Once quantile cut-off points and the universe of cell IDs are generated on the training data, each record that is missing the variable of interest is then assigned into one of the previously determined cell IDs based on its own stratifier values.
2. Next, we generate the posterior probability distribution of the variable of interest for each cell ID using the training data. We use a smooth kernel density approach to generate these distributions.
3. Finally, we drawn *N* random draws from the posterior probability distribution for each cell ID, where *N* is the number of records that are missing the variable of interest within a given cell ID. These random draws are then attached to the records with missing data and used to complete the variable of interest field. [28]
4. After firm birth year is imputed, we add the newly completed records to those that were previously complete and repeat the above steps for firm size.

[26] Stratifiers vary by variable being imputed but include the following: NECF-based age and size, birth year, state, year, number of establishments, whether the firm is located in multiple states, and payroll information.

[27] Note that there is a minimum cell size in place. Once the cell size falls below the minimum, the cell ID "fails" for that run and reverts to the previous stratifier-level.

[28] Since firm size varies from 0 to very large numbers, the firm size smooth kernel density uses size categories ("buckets") for larger firm sizes. When a draw is randomly selected from these buckets, we use a normally distributed random number generator to select the location within the bucket. The largest bucket relies on an exponential distribution with a maximum value.

Horse Race: Methods 1 and 2

In order to assess whether Method 2 generates better imputations than Method 1, we ran two sets of tests. The first set, which we refer to as a macro comparison, identifies which method results in a better approximation to the true distribution. The second set of tests, the micro comparison, tests which method results in imputations closest to the true value at the firm level. The following tests were implemented using a 30% sample of observations from the 2013Q1 FAS_US file that were never missing firm age or size from the BDS. Fifteen percent of these observations were randomly assigned to be missing age and 25% were randomly assigned to be missing size. By using records without missing values for this exercise, we can directly test which edit/imputation strategy yields results that are closer to the truth. Using Method 1 and Method 2, we generated two imputes for each missing variable.[29]

The macro comparison consists of three loss statistics: the Kullback-Leibler (KL) divergence, the Symmetric-Bounded Kullback-Leibler (SBKL) divergence, and the Integrated Mean Square Error (IMSE).[30] Each of these measures the difference between two distributions – in our case the difference between the "true" distribution and the imputed distributions. Specifically, these loss estimates are based on the following equations, where m is the model identifier, i is the evaluation cell (firm age or firm size category), π is the proportion of observations in each evaluation cell using the true distribution, and P is the proportion of observations using the imputed distribution, such that $\sum_i \pi_i = \sum_i P_i = 1$:[31]

$$KL^m = \sum_i \ln\left(\frac{\pi_i^m}{P_i^m}\right)\pi_i^m$$

$$SBKL^m = \sum_i \left[\frac{1}{2}\ln\left(\frac{\pi_i^m}{\frac{1}{2}(\pi_i^m + P_i^m)}\right)\pi_i^m + \frac{1}{2}\ln\left(\frac{P_i^m}{\frac{1}{2}(\pi_i^m + P_i^m)}\right)P_i^m\right]$$

[29] This same 30% sample was also used to identify the stratifiers (and orders of stratifiers) that resulted in the best imputed values of firm age and size.

[30] These tests and code were developed in Sousa's earlier work with John Abowd for LEHD's human capital imputation project.

[31] Missing values (for example, 0's in the denominator or in the ln function) are set to 0.

$$IMSE^m = \sum_i (\pi_i^m - P_i^m)^2$$

Table A4 reports the ratio of Method 2's to Method 1's loss statistics for an easier comparison. It reveals a significant improvement in the similarity of the resulting size and age distributions to the true distributions when we change our methodology from Method 1 to Method 2. All the loss statistics agree that both firm age and firm size distributions are better estimated using Method 2.

Table A4. Loss Statistics for Method 1 and Method 2

	Firm size			Firm age		
	Method 1	Method 2	M2/M1	Method 1	Method 2	M2/M1
IMSE	0.0000227	0.0000017	7.6%	0.0001365	0.0000015	1.1%
KL	0.0002972	0.0000277	9.3%	0.0006652	0.0000154	2.3%
SBKL	0.0000719	0.0000070	9.7%	0.0001959	0.0000041	2.1%

In fact, the new imputation model nearly wipes out any loss associated with missing firm age. However, note that both methods yield very small loss estimates. Though the decrease in size of the loss statistics when comparing Method 1 to Method 2 is dramatic, closer inspection of the resulting distributions reveal that the two methods actually generate very similar distributions. That is, though the "losses" associated with Method 1 are higher than those associated with Method 2, they are still fairly small. Table A5 shows the similarity between the three distributions.

Table A5. Distributions of Firm Age and Size, True and Imputed

Firm Size	Truth	Method 1	Method2
0-19	86.8	87.2	86.7
20-49	7.4	7.3	7.4
50-249	3.9	3.9	3.9
250-499	0.5	0.5	0.6
500+	1.4	1.1	1.5
Firm Age	Truth	Method 1	Method 2
Inactive	0.0	0.1	0.0
0-1 years	10.8	11.1	10.8
2-3 years	14.4	14.7	14.4
4-5 years	11.2	11.4	11.2
6-10 years	20.0	20.2	20.0
11+ years	43.6	42.6	43.7

The results from the macro analysis suggest that Method 2 does a superior job in replicating the overall age and size distributions than Method 1, particularly in the extreme size and age cells. However, both approaches yield distributions that are very close to the true distributions.

Since LEHD data products report labor dynamics at small geographic, demographic, and industrial cells, we also need to consider which strategy does a better of job of predicting actual firm age and size at the firm level. The micro comparison does exactly this by comparing the true firm size and age values to the imputed values for each method. Tables A6 and A7 report the proportion of firms in each age and size group that were imputed into the correct age or size group along the diagonal, while off-diagonal values are the proportion of firms that were misallocated to each size or age group. For example, the top row shows that 99.4% of records that had a true size of 0 to 19 workers were correctly allocated into the 0-19 workers category by Method 1 while another 0.5% were allocated into the 20 to 49 group.

Table A6. Proportion of Each Firm Size Category, by Firm Size Group of Imputation (Imputed Observations Only)

	Method 1				
Truth	0-19	20-49	50-249	250-499	500+
0-19	99.4	0.5	0.0	0.0	0.0
20-49	17.0	80.9	2.1	0.0	0.0
50-249	10.5	12.4	76.4	0.6	0.1
250-499	16.6	9.9	28.2	43.4	1.9
500+	23.0	13.4	27.8	10.4	25.4
	Method 2				
Truth	0-19	20-49	50-249	250-499	500+
0-19	95.8	2.5	0.8	0.2	0.7
20-49	30.3	56.9	8.8	0.9	3.2
50-249	13.9	18.7	53.7	4.1	9.7
250-499	18.3	10.5	30.3	23.3	17.6
500+	24.0	12.9	23.8	8.9	30.4

There are a couple of important things to note from Table A6. First, using NECF size to replace missing firm size values, which is what Method 1 does, does a great job of correctly identifying smaller firms – particularly those with less than 50 workers. The only group for which Method 2 is better at identifying firm size is for the firms with over 500 employees. Note that less than 6% of all jobs are in firms with 250-499 employees, hence the low rate of correct imputation for this size group is not particularly worrisome. However, NECF size is systematically lower than BDS size: that is, it is almost never the case that NECF size predicts a firm size that is larger than the true size.[32] Instead, it is systematically biased downwards. Method 2, on the other hand, is less likely to correctly impute the true size category for smaller firms. However, since nearly half of all jobs are in firms with over 500 employees, correctly predicting the large size category correctly is very important. Method 2 also correctly identifies firms in the smallest size group, which account for 19% of all jobs. Another consideration is that Method 2 does not have the same obvious downward bias as Method 1, which is why the macro comparison revealed that Method 2's aggregate distribution fits the truth better than Method 1's.

[32] Recall that the true size is March 12th employment from the previous year while the preferred NECF size is March employment from the previous year.

Table A7. Proportion of Each Firm Age Category, by Firm Age Group of Imputation (Imputed Observations Only)

	Method 1					
Truth	Inactive	0-1	2-3	4-5	6-10	11+
Inactive	99.4	0.0	0.3	0.3	0.0	0.0
0-1 years	1.3	90.3	3.0	1.1	1.6	2.8
2-3 years	0.6	6.6	82.4	2.6	3.0	4.9
4-5 years	0.4	4.1	7.9	72.8	6.2	8.8
6-10 years	0.3	3.7	5.4	7.2	65.9	17.6
11+ years	0.2	2.9	4.4	5.2	15.8	71.5
	Method 2					
Truth	Inactive	0-1	2-3	4-5	6-10	11+
Inactive	4.1	10.6	18.1	16.3	26.9	24.1
0-1 years	0.2	68.1	14.3	3.6	5.1	8.8
2-3 years	0.1	12.9	58.9	10.0	6.9	11.3
4-5 years	0.0	2.9	13.6	51.2	15.6	16.7
6-10 years	0.0	2.3	4.3	8.8	55.0	29.6
11+ years	0.0	1.8	3.2	4.1	14.0	76.9

Table A7 reports the same analysis conducted on all observations where age was imputed (where Method 1's primary imputation strategy is relying on NECF age). Once again, Method 1 does a superior job of correctly predicting all but the oldest firm age group. Unlike with the size imputations, Method 1's age impute is not as obviously biased downwards as its size impute, with plenty of observations being assigned an age older through Method 1 than their true age based on the BDS. Further analysis reveals that Method 1's success is only due to the records that were imputed with NECF-based age, not to the left-censored/successor-predecessor firms that were imputed. Table A8 shows just the values of the diagonals from A7 when the data are limited to left-censored/successor-predecessor firms. For that subset of firms, Method 2 is a better approach, doing a superior job of correctly predicting young firms and older firms.

Table A8. Proportion of Each Firm Age Category, by Firm Age Group of Imputation (Left-censored/Successor-Predecessor Observations Only)

	Method 1	Method 2
0-1 years	9.8	14.3
2-3 years	14.9	15.5
4-5 years	14.2	14.0
6-10 years	28.2	27.3
11+ years	78.8	80.7

With the exception of the largest/oldest groups, this micro analysis reveals that Method 1 tends to more accurately predict firm age and size than Method 2. This is especially the case when identifying young firms: using NECF-based firm birth (Method 1) is superior to imputing firm birth. Both strategies result in negatively biased size imputes on average, though the largest/oldest firms are more likely to be correctly identified using Method 2.

Method 3

After considering the results of the macro and micro analyses, we decided to test a third approach, Method 3. This approach is identical to Method 2 for firm size, but combines Methods 1 and 2 for firm age. That is, Method 3 utilizes Method 2's cell ID code to impute firm size for

Table A9. Loss Statistics For Method 3, and Comparison to Method 1

	Firm size		Firm age	
	Method 3	M3/M1	Method 3	M3/M1
IMSE	0.0000022	9.6%	0.0001129	82.7%
KL	0.0000336	11.3%	0.0006418	96.5%
SBKL	0.0000085	11.8%	0.0001909	97.5%

all firms missing size in their first observation and firm age for those firms that are left-censored in the NECF or associated with a successor-predecessor event. The loss estimates for Method 3 are presented in Table A9 while Table A10 reports the corresponding distributions. Note that the loss estimates for firm size increased slightly from Method 2 though the same imputation model was used. This is because firm age is imputed first and is then used as one of the stratifiers to build the posterior probability distribution. However, Table A10 clearly shows that the distributional changes were minimal – once rounded to nearest tenth, the distributions of

Methods 2 and 3 are the same. The loss estimates for firm age are lower than Method 1's and higher than Method 2's; this is as expected since Method 3 is a combination of the first two.

Table A10. Distributions of Firm Age and Size, True and Imputed Using Method 3

Firm Size	Truth	Method 3
0-19	86.8	86.7
20-49	7.4	7.4
50-249	3.9	3.9
250-499	0.5	0.6
500+	1.4	1.5
Firm Age	**Truth**	**Method 3**
Inactive	0.0	0.1
0-1 years	10.8	11.2
2-3 years	14.4	14.7
4-5 years	11.2	11.3
6-10 years	20.0	20.1
11+ years	43.6	42.7

Since the results changed so little for the firm size imputations using Method 3, Table A11 only reports the micro comparison for the firm age imputes. Using the NECF-based age for missing age records of FAS_EINs for which birth is observed and then imputing age only for the left-censored/successor-predecessor firms results in over 90% accurate identification of new firms and over 70% accurate identification of the oldest category of firms. As a quick comparison to Table A7 confirms, this mixed strategy successfully combines the best of Methods 1 and 2.

Table A11. Proportion of Each Firm Age Category, by Firm Age Group of Imputation for Method 3 (Imputed Observations Only)

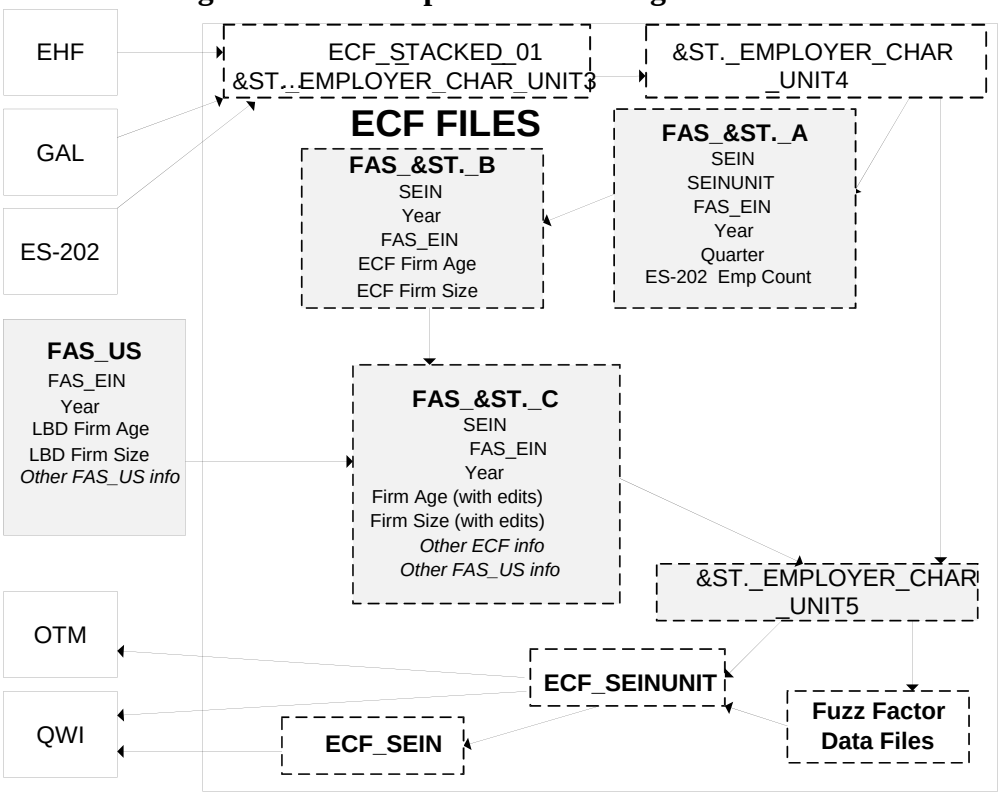
Truth	Inactive	0-1	2-3	4-5	6-10	11+
Inactive	99.4	0.0	0.3	0.3	0.0	0.0
0-1 years	1.3	90.5	3.1	1.0	1.5	2.5
2-3 years	0.6	6.6	82.5	2.7	2.9	4.8
4-5 years	0.4	4.0	7.8	72.7	6.1	9.0
6-10 years	0.3	3.6	5.2	7.0	65.6	18.3
11+ years	0.2	2.8	4.2	5.0	15.0	72.8

Appendix IV

Modifications to the ECF code to incorporate the FAS_US file

The ECF enhancements to accommodate the integration of firm age and size begin by reading in the fourth dataset in the "employer_char_unit" sequence, as shown in Figure A1. First, the FAS_EIN variable is created using the EIN and the rules described above. Next, state-specific firm birth year, birth quarter, and size are calculated at the FAS_EIN level. Then, the ECF records are merged with the FAS_US data and FAS_US birth year, birth quarter, and size are applied whenever available. When an element is missing, FAS_US birth year and quarter are sometimes forecasted. Otherwise, data based on the state ECF are used. This process is described in detail in this section.

Figure A1: State-Specific Processing Data Flow Overview



Recall that the ECF process is at the state level, meaning data from only one state is used, while the NECF is national but uses the previous quarter's data submissions. For the case in which a state's resubmission of data includes new EINS, state-level age and size information from the

state's ECF are used to complete firm age and size. As in the national process, birth year and quarter are assigned based on when they first appear in the ECF. Size is also calculated at this level in the same manner as in the FAS_US file. After the ECF process generates state-specific birth year, quarter, and size information, this file is merged to the FAS_US file. Only firms that do not match to the FAS_US file, those that were never included in a state's data submissions in prior quarters, will rely on the ECF-based process for firm age and size.[33] Firm size requires a positive match to the FAS_US file at the FAS_EIN and year level. Firm age will accept a lagged match of up to two years: that is, if a firm does not match to the FAS_US file in a given year, the matching algorithm searches for an FAS_US record up to 2 years into the future. After all data have been completed at the FAS_EIN and year level, the data are merged to the quarterly-SEINUNIT level ECF data (&st_employer_char_unit5). It is only at this point that the firm birth year and quarter data are transformed into firm age (since firm age varies quarterly). Firm age is reported in years. As mentioned above, the quarter of birth variable is not a research-quality variable; it has been introduced specifically to smooth the birth distribution over the year.

[33] In the unlikely event that no new FAS_US file is produced in time for the QWI production run, the previous quarter's FAS_US file will be used.

Appendix V

QWI Confidentiality Protection Procedures and Disclosure Avoidance

Released QWI data utilize a combination of noise infusion, item suppression, weighting and multiple imputation of missing data to ensure the confidentiality of all establishment-level micro-data used for tabulating indicators about employers and employees. The nature of the protection mechanism ensures that additional detail in both the number of characteristics and categories used in tabulation does not compromise confidentiality. These protection measures are described in detail in Abowd, Stephens and Vilhuber (2006) and are outlined in this section for completeness. The structure of the existing disclosure avoidance protocol will be applied to the new QWI tabulations that include firm age and size, with the addition of one additional noise factor described in detail below. The existing fuzz-factor database will be used for all tabulations. Missing data will be multiply imputed (10 implicates) for the entire universe. Hence, the new tabulations will be based on the same set of fuzzed, weighted, and multiply imputed establishment-level microdata as other tabulations produced for the QWI.

Dynamically Consistent Noise Infusion

The QWI's multiplicative noise infusion procedure involves the following steps:

1. Two sets of random fuzz factors are generated. One set applies to state-level Unemployment Insurance accounts (SEIN, employers) and another applies to employment establishments (SEINUNIT), of which there can be several per employer. Each fuzz factor has a minimum and maximum amount of distortion: they can be greater than or less than one, but a confidential range around the value 1 is excluded. All establishments (SEINUNIT) owned by the same employer (SEIN) have fuzz factors on the same side of one.

2. Particular formulas apply to subsets of each of the thirty economic indicators for each employment establishment. For most measures, such as beginning of quarter employment, the microdata for the indicator is multiplied by the distortion factor in order to produce the distorted indicator at the establishment level that is used for subsequent tabulation. Other measures apply the fuzz factors in other manners consistent with the

nature of the indicator. This process is described in detail in Abowd, Stephens and Vilhuber (2006).

Item Suppression

The noise infusion procedure described above is not sufficient to protect confidentiality of very small released cells, defined by a confidential combination of employer characteristics, geography, and employee demographic characteristics. To supplement the noise infusion, the following process is employed:

1. The process checks whether the source data are available for a particular quarter, and, if they are not, applies a flag to released data indicating that the element is missing (status flag values -2 and -1).
2. If the data element fails to meet U.S. Census Bureau publication standards for the handling of small cells subject to noise infusion, the element is not released and a flag is added to the released data (status flag value 5).
3. After that, a separate flag is applied if the distortion to a particular cell exceeds the limit set by the U.S. Census Bureau Disclosure Review Board, and a distortion flag is applied to the released data (status flag value 9).
4. True zeros are noted (status flag 0). This results in missing data for some ratios.
5. Minimally distorted data are noted (status flag 1).

At the end of this process, items with status flags 0, 1, and 9 are released. In all cases, indicators are computed from the distorted data. Items with status flags -2, -1, and 5 are set to "missing data" in the release file.

Weighting

The QWI production process includes an establishment-level weight that measures the proportionate difference between a particular QWI employment indicator (beginning of quarter employment) and an employer-level total reported employment variable. The application of this weight provides an additional degree of confidentiality to the released QWI data.

Missing Data Imputation

Missing demographic and workplace data are multiply-imputed using a combination of empirical Bayes procedures documented in Abowd et al. (2009). The existing documentation covers age, sex, employer geography, and industry imputation. Note that workplace characteristics are multiply-imputed via the Unit-to-Worker imputation described in the paper cited here. The procedure for imputing missing race, ethnicity and education data is an enhancement of the procedure described in that paper. Missing data rates vary substantially. Age and sex are missing for approximately 4% of the universe. Race and ethnicity are partially missing for about 17% of the universe, though there are ancillary data for most of the partial missing. Educational attainment is missing for about 90% of the universe. Workplace characteristics (geography and industry) are missing for between 30% and 40% of the universe. Multiple imputation of missing data also contributes to confidentiality protection since the imputed characteristics are estimates of the missing values, not the confidential values themselves. Note that, currently, firm age and size are not multiply imputed.

New Application of Noise Infuser Parameter for Firm Size

The enhancement to the confidentiality protection procedure for the new tabulations is a straightforward application of the current confidentiality protection measures applied to the economic indicators. The only challenge is that one of the desired variables by which cross-tabulations will be made, firm size, is strongly correlated with several measures that are currently released as economic indicators: establishment-level employment counts. To ensure that releasing data by firm size does not allow users to recover the confidential noise infusion parameters, firm size will be multiplied by the noise infusion parameter that applies to the business's UI account number (SEIN) as follows:

$$firmsize_{fuzz} = rampSEIN \quad * firmsize$$

Note that a firm that is associated with multiple UI account numbers (SEIN) will have its firm size multiplied by a different noise infusion value in each state in which it does business, creating multiple values of firm size that will be used for tabulation.

Appendix VI

Application to OnTheMap

Firm age and firm size data are included in OnTheMap data releases beginning with the OnTheMap 2012 data release. In order to avoid disclosure related to firm age and size, these variables have a limited release in OnTheMap. For one, they will be made available only in the Work Area Characteristics (WAC) tables. Unlike many other characteristics, firm age and size will not be crossed with any other variables in the WAC tables. Instead, firm age and firm size will be available for only the total number of jobs in the geographic area under analysis. Attempting to subset these results by demographic characteristics, for example, results in firm age and size being dropped from the results.